

Variational Inference for Bayesian Computation: An Application to NBA Salary Data

Research Seminar in Selected Topics in Statistical
Learning and Data Science

Adrian Wimmer | 18 June 2026



1. Introduction

What is Variational Bayes?

2. Variational Bayes in Action

Bayesian Linear Regression Model

CAVI Updates

MCMC Benchmark

3. Simulation Study

4. Application: NBA Salary Data

5. Conclusion

Introduction

In Bayesian statistics the central object is the **posterior**

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathbf{y})}.$$

The marginal likelihood

$$p(\mathbf{y}) = \int p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}$$

is **not always available in closed form**, making exact inference infeasible.

Two widely used approximate methods:

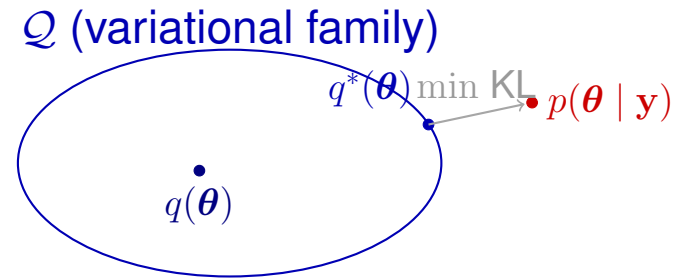
- **MCMC** (e.g. Gibbs sampling) — accurate but computationally expensive, especially for large n or p .
- **Variational Bayes (VB)** — turns inference into an optimisation problem; much faster, quality depends on the chosen family.

What is Variational Bayes?

Problem. The true posterior $p(\boldsymbol{\theta} \mid \mathbf{y})$ is intractable.

Idea. Approximate it by the closest member of a tractable family $\mathcal{Q}$, measured by KL divergence:

$$q^*(\boldsymbol{\theta}) = \arg \min_{q \in \mathcal{Q}} \text{KL}[q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta} \mid \mathbf{y})].$$



Variational Bayes in Action

Bayesian Linear Regression Model

Data model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

Likelihood

$$p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right\}.$$

Priors. We place independent conjugate priors on $\boldsymbol{\beta}$ and σ^2 (1):

$$\boldsymbol{\beta} \sim \mathcal{N}_p(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta), \quad \sigma^2 \sim \text{IG}(A, B),$$

with densities

$$p(\boldsymbol{\beta}) = (2\pi)^{-p/2} |\boldsymbol{\Sigma}_\beta|^{-1/2} \exp\left\{-\frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta)^\top \boldsymbol{\Sigma}_\beta^{-1}(\boldsymbol{\beta} - \boldsymbol{\mu}_\beta)\right\},$$

$$p(\sigma^2) = \frac{B^A}{\Gamma(A)} (\sigma^2)^{-(A+1)} \exp\left\{-\frac{B}{\sigma^2}\right\}, \quad \sigma^2 > 0.$$

Mean-Field Variational Bayes

Core idea. Choose a family $\mathcal{Q}$ and find

$$q^*(\boldsymbol{\theta}) = \arg \min_{q \in \mathcal{Q}} \text{KL}[q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta} \mid \mathbf{y})].$$

Mean-field factorisation

$$q(\boldsymbol{\beta}, \sigma^2) = q(\boldsymbol{\beta}) q(\sigma^2).$$

Assumes $\boldsymbol{\beta}$ and σ^2 are *independent* under q .

Coordinate updates follow from the expected log-joint:

$$\begin{aligned} \log q(\boldsymbol{\beta}) &\propto \mathbb{E}_q[\log p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2) + \log p(\boldsymbol{\beta})], \\ \log q(\sigma^2) &\propto \mathbb{E}_q[\log p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2) + \log p(\sigma^2)]. \end{aligned}$$

Recognising the kernel of each factor:

$$q(\boldsymbol{\beta}) = \mathcal{N}_p(\boldsymbol{\mu}_{q(\boldsymbol{\beta})}, \boldsymbol{\Sigma}_{q(\boldsymbol{\beta})}) , \quad q(\sigma^2) = \text{IG}(A_{q(\sigma^2)}, B_{q(\sigma^2)}) .$$

CAVI Updates

Iterating the coordinate updates gives the **CAVI algorithm**.

Shape parameter

$$A_{q(\sigma^2)} = A + \frac{n}{2}.$$

Scale parameter

$$B_{q(\sigma^2)} = B + \frac{1}{2} \left[(\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_{q(\beta)})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_{q(\beta)}) + \text{tr}(\mathbf{X}^\top \mathbf{X} \boldsymbol{\Sigma}_{q(\beta)}) \right].$$

Variational covariance

$$\boldsymbol{\Sigma}_{q(\beta)} = \left(\frac{A_{q(\sigma^2)}}{B_{q(\sigma^2)}} \mathbf{X}^\top \mathbf{X} + \boldsymbol{\Sigma}_\beta^{-1} \right)^{-1}.$$

Variational mean

$$\boldsymbol{\mu}_{q(\beta)} = \boldsymbol{\Sigma}_{q(\beta)} \left(\frac{A_{q(\sigma^2)}}{B_{q(\sigma^2)}} \mathbf{X}^\top \mathbf{y} + \boldsymbol{\Sigma}_\beta^{-1} \boldsymbol{\mu}_\beta \right).$$

Fixed: $\mathbf{X}, \mathbf{y}, \boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, A, B$ and $A_{q(\sigma^2)}$. **Updated per iteration:** $B_{q(\sigma^2)}, \boldsymbol{\Sigma}_{q(\beta)}, \boldsymbol{\mu}_{q(\beta)}$.

Introduction



Variational Bayes in Action



Simulation Study



Application: NBA Salary Data



Conclusion

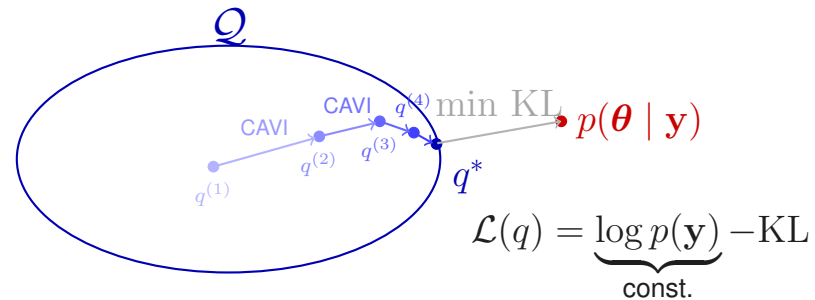


Literatur

Evidence Lower Bound (ELBO)

How do we measure progress? Minimising KL directly is intractable — instead, CAVI maximises the **ELBO**:

$$\mathcal{L}(q) = \log p(\mathbf{y}) - \text{KL}[q(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta} \mid \mathbf{y})].$$



Maximising $\mathcal{L}(q) \Leftrightarrow$ minimising $\text{KL}[q \parallel p]$, since $\log p(\mathbf{y})$ is constant in q .

Evidence Lower Bound (ELBO)

CAVI maximises the **ELBO**

$$\begin{aligned}\mathcal{L}(q) = & \underbrace{\mathbb{E}_q[\log p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2)]}_{\text{Likelihood}} + \underbrace{\mathbb{E}_q[\log p(\boldsymbol{\beta})]}_{\boldsymbol{\beta}\text{-Prior}} + \underbrace{\mathbb{E}_q[\log p(\sigma^2)]}_{\sigma^2\text{-Prior}} \\ & - \underbrace{\mathbb{E}_q[\log q(\boldsymbol{\beta})]}_{\boldsymbol{\beta}\text{-Entropy}} - \underbrace{\mathbb{E}_q[\log q(\sigma^2)]}_{\sigma^2\text{-Entropy}}.\end{aligned}$$

ELBO — Explicit Expression

Evaluating each term yields the closed-form expression:

$$\begin{aligned}\mathcal{L}(q) = & \underbrace{-\frac{n}{2} \log(2\pi) - \frac{n}{2} [\log B_{q(\sigma^2)} - \psi(A_{q(\sigma^2)})] - \frac{1}{2} \frac{A_{q(\sigma^2)}}{B_{q(\sigma^2)}} [(\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_{q(\beta)})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_{q(\beta)}) + \text{tr}(\mathbf{X}^\top \mathbf{X} \boldsymbol{\Sigma}_{q(\beta)})]}_{\text{Likelihood term}} \\ & + \underbrace{-\frac{p}{2} \log(2\pi) - \frac{1}{2} \log |\boldsymbol{\Sigma}_\beta| - \frac{1}{2} [\text{tr}(\boldsymbol{\Sigma}_\beta^{-1} \boldsymbol{\Sigma}_{q(\beta)}) + (\boldsymbol{\mu}_{q(\beta)} - \boldsymbol{\mu}_\beta)^\top \boldsymbol{\Sigma}_\beta^{-1} (\boldsymbol{\mu}_{q(\beta)} - \boldsymbol{\mu}_\beta)]}_{\beta\text{-Prior term}} \\ & + \underbrace{A \log B - \log \Gamma(A) - (A + 1) [\log B_{q(\sigma^2)} - \psi(A_{q(\sigma^2)})] - B \frac{A_{q(\sigma^2)}}{B_{q(\sigma^2)}}}_{\sigma^2\text{-Prior term}} \\ & + \underbrace{\frac{1}{2} \log |\boldsymbol{\Sigma}_{q(\beta)}| + \frac{p}{2} \{1 + \log(2\pi)\}}_{\beta\text{-Entropy term}} + \underbrace{A_{q(\sigma^2)} + \log B_{q(\sigma^2)} + \log \Gamma(A_{q(\sigma^2)}) - (A_{q(\sigma^2)} + 1) \psi(A_{q(\sigma^2)})}_{\sigma^2\text{-Entropy term}}\end{aligned}$$

Fixed: $\mathbf{X}, \mathbf{y}, \boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, A, B$. **Updated:** $B_{q(\sigma^2)}, \boldsymbol{\Sigma}_{q(\beta)}, \boldsymbol{\mu}_{q(\beta)}$. $\psi(x) = \frac{d}{dx} \log \Gamma(x)$: Digamma function.

MCMC as a Computational Benchmark

Although the model is relatively simple, MCMC can become expensive for large n or p . The joint posterior is

$$p(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}, \mathbf{X}) \propto p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2, \mathbf{X}) p(\boldsymbol{\beta}) p(\sigma^2).$$

Gibbs sampling draws iteratively from the **full conditionals**:

Full conditional for $\boldsymbol{\beta}$

$$p(\boldsymbol{\beta} \mid \sigma^2, \mathbf{y}, \mathbf{X}) = \mathcal{N}_p(\boldsymbol{\mu}_\beta^*, \boldsymbol{\Sigma}_\beta^*)$$
$$\boldsymbol{\Sigma}_\beta^* = (\sigma^{-2} \mathbf{X}^\top \mathbf{X} + \boldsymbol{\Sigma}_\beta^{-1})^{-1}, \quad \boldsymbol{\mu}_\beta^* = \boldsymbol{\Sigma}_\beta^* \left(\sigma^{-2} \mathbf{X}^\top \mathbf{y} + \boldsymbol{\Sigma}_\beta^{-1} \boldsymbol{\mu}_\beta \right).$$

Full conditional for σ^2

$$p(\sigma^2 \mid \boldsymbol{\beta}, \mathbf{y}, \mathbf{X}) = \text{IG}(A^*, B^*), \quad A^* = A + \frac{n}{2}, \quad B^* = B + \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

This motivates VB as a **deterministic alternative**: instead of sampling, approximate the posterior by a simpler distribution.

Simulation Study

We compare CAVI and Gibbs sampling in the Bayesian linear regression model for increasing problem sizes (n, p) , while keeping the ratio p/n fixed.

Here, p denotes the number of regression coefficients including the intercept.

n	p	CAVI (s)	Gibbs (s)	Speed-up
100	10	< 0.01	0.19	> 19×
1 000	100	0.14	1.70	12.1×
5 000	500	16.05	151.15	9.4×
10 000	1 000	128.53	1 414.59	11.0×

- The speed-up is measured as Gibbs time/CAVI time.
- In the largest setting, CAVI is roughly 12× **faster**.
- CAVI requires about 129 s, while Gibbs sampling takes about 1,415 s, or roughly 23.6 minutes, for 1 000 iterations.

Application: NBA Salary Data

Motivation. Inspired by the *Moneyball* idea:
Which on-court statistics actually drive NBA salaries — and can we identify undervalued players?

Basketball produces rich **box score data** for every player and game..

Response	log salary, then standardised
Observations	$n = 7,298$ player-season records, 2010–2025
Predictors	$p = 19$ including intercept

Category	Predictors
Season / profile	Year, Age
Playing time / role	G = games played, GS = games started, MP = minutes played
Shooting	FG = field goals made, FGA = field goal attempts, eFG% = effective field goal percentage
Free throws	FT = free throws made, FTA = free throw attempts, FT% = free throw percentage
Rebounding	TRB = total rebounds
Playmaking / defense	AST = assists, STL = steals, BLK = blocks
Ball security / discipline	TOV = turnovers, PF = personal fouls
Scoring output	PTS = points scored

Model Specification

We fit the Bayesian linear regression model on **standardised log-salary** and **standardised predictors**:

$$y_i^{\text{scaled}} = \frac{\log(S_i) - \bar{y}}{s_y} = \beta_0^{\text{scaled}} + \sum_{j=1}^p \beta_j^{\text{scaled}} \frac{x_{ij} - \bar{x}_j}{s_{x_j}} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

Why log-salary? Raw salaries are highly skewed; modelling $\log(S_i)$ is more appropriate for a Gaussian linear regression model.

Why standardisation? Standardisation puts all predictors on a comparable scale and improves numerical stability of CAVI.

Priors on the scaled model parameters:

$$\beta^{\text{scaled}} \sim \mathcal{N}(\mathbf{0}, 100I), \quad \sigma^2 \sim \text{IG}(1, 1).$$

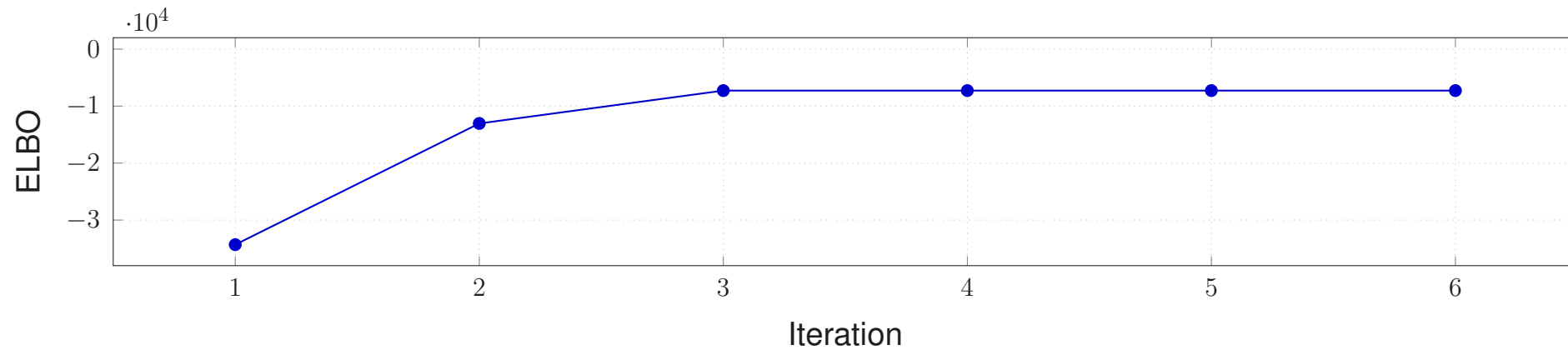
Back-transformation to the original log-salary scale:

$$\beta_j^{\text{orig}} = \beta_j^{\text{scaled}} \frac{s_y}{s_{x_j}}, \quad j = 1, \dots, p-1.$$

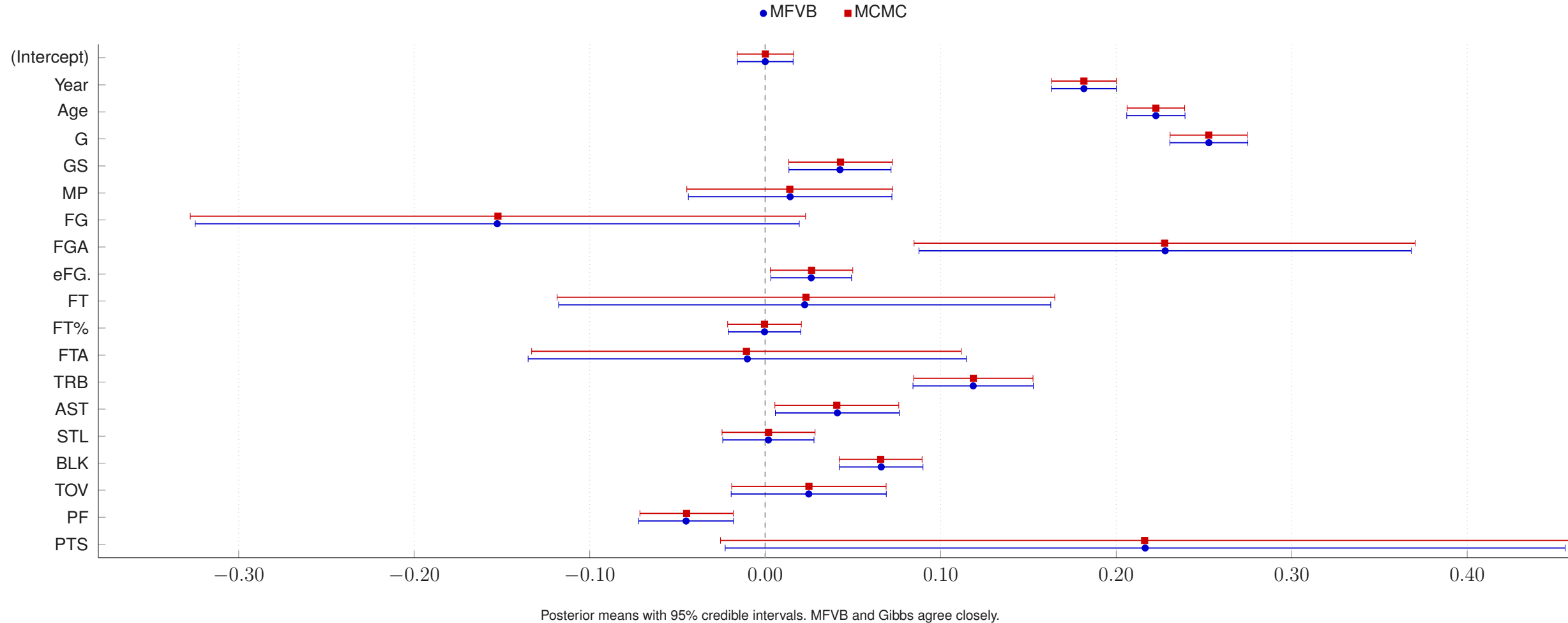
Thus, β_j^{orig} describes the change in $\log(\text{salary})$ for a one-unit increase in the original predictor x_j .

CAVI Convergence

The ELBO increases monotonically and converges after **5 iterations**:



Results: Standardised Posterior Coefficients



Introduction

○○

Variational Bayes in Action

○○○○○○○

Simulation Study

○

Application: NBA Salary Data

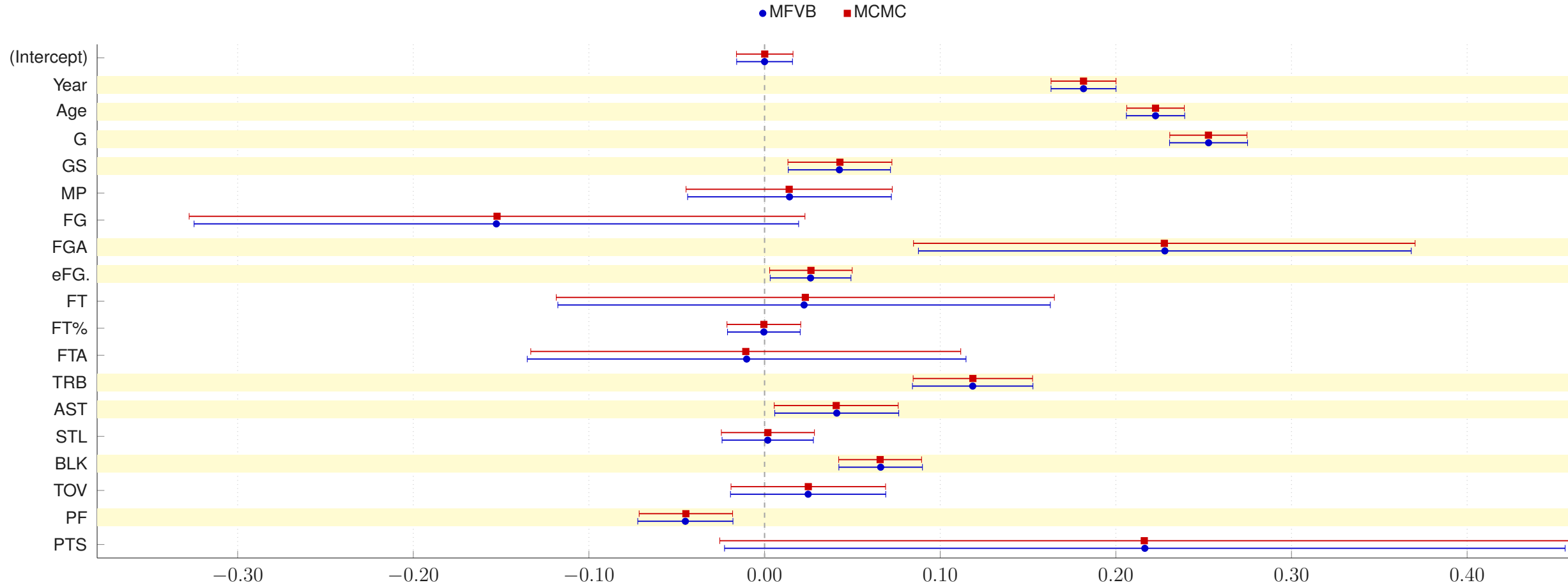
○○○●○○

Conclusion

○

Literatur

Results: Standardised Posterior Coefficients



Highlighted rows indicate coefficients whose 95% credible intervals do not include zero.

Introduction

○ ○

Variational Bayes in Action

○ ○ ○ ○ ○ ○ ○ ○

Simulation Study

○

Application: NBA Salary Data

○ ○ ○ ○ ● ○ ○

Conclusion

○

Literatur

Interpretation on the Original Scale

For standardised response and predictors, the coefficients on the original scale are

$$\beta_j^{\text{orig}} = \beta_j^{\text{scaled}} \frac{s_y}{s_{x_j}}, \quad j = 1, \dots, p - 1.$$

A change of Δx_j units implies the percentage effect

$$100 \left(\exp \left(\beta_j^{\text{orig}} \Delta x_j \right) - 1 \right) \%.$$

Variable	Unit change	MFVB (%)	MCMC (%)
Year	+1 raw unit	5.74	5.74
Age	+1 raw unit	7.74	7.74
G	+1 raw unit	1.52	1.52
GS	+1 raw unit	0.22	0.22
FGA	+1 raw unit	7.23	7.22
eFG.	+1 percentage point	0.39	0.39
TRB	+1 raw unit	7.18	7.18
AST	+1 raw unit	3.22	3.20
BLK	+1 raw unit	24.72	24.59
PF	+1 raw unit	-8.20	-8.14

Percentage effects on salary, holding all other predictors constant. For eFG., the effect refers to a one percentage point increase.

Conclusion

Main findings

- Mean-field VB provides a deterministic and computationally efficient alternative to MCMC.
- CAVI is substantially faster than Gibbs sampling for larger problem sizes.
- In the NBA application, MFVB and MCMC lead to almost identical posterior coefficient estimates.

Possible extensions

- *Dataset*: Identify potentially underpaid players and examine whether their next contracts adjust upward.
- *Method*: Use richer variational families, stochastic variational inference, or more flexible model structures.
- *Application*: Include advanced basketball statistics to improve salary prediction.

Introduction



Variational Bayes in Action



Simulation Study



Application: NBA Salary Data



Conclusion



Literatur

References I

- [1] Gelman, A. et al. (2013). *Bayesian Data Analysis*. Chapman and Hall/CRC.
- [2] Ormerod, J. T. and Wand, M. P. (2010). Explaining variational approximations. *The American Statistician*.
- [3] Casella, G. and George, E. I. (1992). Explaining the Gibbs Sampler. *The American Statistician*.
- [4] Ratin21 (n.d.). NBA Player Stats and Salaries 2010–2025. Kaggle.

Introduction



Variational Bayes in Action



Simulation Study



Application: NBA Salary Data



Conclusion



Literatur